

Biopharma Technical Assessment

IT Developer
Ful



ACCELERATE SUSTAINABLE GROWTH



Mini Project

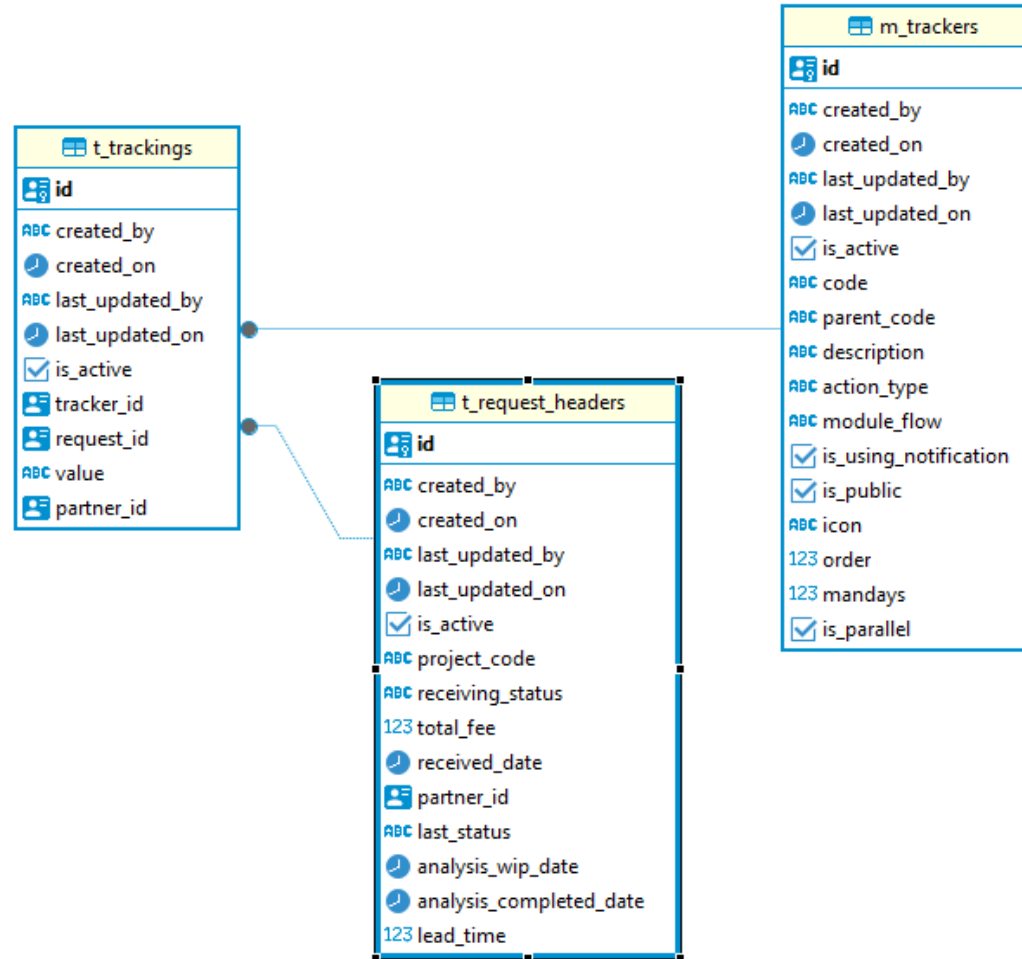
Preparations

- Coding environment tool.
- Notes (if needed).

Criteria

- Solve the issue using .NET or Next JS.
- Solve the issue within the mentioned duration of time.
- Scoring priority:
 1. Functional
 2. UI & UX
 3. Security
 4. Unit/Integration Tests
 5. Logging
 6. QoL

Database - Diagram



Issue

- Based on Database's diagram, create a web app for CRUD (create, read, update, delete) based on these designs:
(Note: You can improve the designs as needed)
- Features:
 1. Auth
 2. CRUD Master Trackers
 3. CRUD Transaction Requests
- 4. Logging (create your own backend using .NET)
- 5. CRUD Transaction Trackings (bonus)

Master - Trackers



MAIN MENU

Dashboard

Request

Tracking Progress

Data History

Documents

USER

Profile

Change Password

MASTER

Customer

Tracker

Document

Notification

Setup

Lab Analysis

Master Tracker

Manage tracker data

Show 10

Search: Search Item

Add

Action	Code	Parent Code	Description	Order	Icon	Action Type	Module Flow	Mandays	Parallel	Public	Notification
...	REQ-CREATE		Created Request	0		none	CA	-	X	✓	X
...	REQ-SUBMIT	REQ-CREATE	Submitted Request	1		none	CA	-	X	✓	X
...	REQ-CONFIRM	REQ-SUBMIT	Confirmed Request	2		text	CA	3	X	✓	X
...	UPLOAD-QUOT	REQ-SUBMIT	Quotation Uploaded	2		file	CA	0	X	✓	X
...	REQ-ACC	REQ-CONFIRM	Accepted Request	3		none	CA	-	X	✓	X
...	UPLOAD-PO	UPLOAD-QUOT	PO Uploaded	3		file	CA	0	X	✓	X
...	SAMPLE-SENT	REQ-CREATE	Sample Sent	4		date	CA	-	X	✓	X
...	RFA-PROJECT-CODE	REQ-CREATE	RFA Project Code	4		text	CA	-	X	✓	X
...	SAMPLE-RECEIVE	SAMPLE-SENT	Sample Received	5		date	CA	-	X	✓	X
...	ANL-WIP	SAMPLE-RECEIVE	Analysis In Progress	6		date	CA	3	X	✓	X

Showing 1 to 10 of 18 entries

Prev 1 Next

Transaction - Requests



MAIN MENU

Dashboard

Request

Lab Analysis

BA/BE

Tracking Progress

Data History

Documents

USER

Profile

Change Password

MASTER

Customer

Tracker

Document

Notification

Setup

Lab Analysis

PharmaMetricLabs

Lab Analysis Request

Manage all data

Show 10 Filter By Add RFA List

Action	Request ID	Company Name	Project Code	Request Date	Last Status	Receiving Status	Total Fee
...	BFECE3AB41AD45229A0F...	PT KALBE FARMA TBK	CA20250620133744	20 Jun 2025	Not sent yet	Not received yet	Rp 1,00
...	5B110F30AA494B4EBE827...	PT KALBE FARMA TBK	PRJ-250623	20 Jun 2025	RFA Project Code	Sample was sent on 2025-06-23	Rp 400.000,00

Showing 1 to 2 of 2 entries



API Docs - Basic Information

- Base URL:
<https://api-assessment-biopharma.kalbe.co.id>
- Auth Path:
/api/v1/authentication
- Main Features Path:
/api/v1/feature
- Main Features Docs:
[Postman](#)

API Docs - Auth

- App Auth Headers:
app_id=41e4f05a
app_key=306db443aa95cecd671214b806bb31ef
- Auth Headers:
Authorization=Bearer {access_token}
- Refresh Token: pass back the refresh token cookie from the login response to get the new access token
- All of endpoints need Authorization with access bearer token passed except for log in

API Docs - Auth

- Log In: [POST] /auth/login

```
{  
  "appCode": "ASSESSMENT",  
  "moduleCode": "ALL",  
  "user": "assessment.it",  
  "password": "Password123?"  
}
```

- Log Out: [POST] /auth/logout

```
{  
  "appCode": "ASSESSMENT",  
  "moduleCode": "ALL",  
  "user": "assessment.it"  
}
```

- Refresh Token: [GET] /auth/token/refresh?refreshAuthorization=true
* “refreshAuthorization” param can be adjusted with true or false as needed
- Authorization (Roles & Modules): [GET] /auth/authorization?nested=false
* “nested” param can be adjusted with true or false as needed

API Docs – Paginations

- Simple Pagination Headers:

X-Paging-Offset: (long) Start index (e.g., 0)

X-Paging-Limit: (long) Max number of records (e.g., 25)

X-Paging-SortBy: (string) Property name to sort by (e.g., trackerCode or converted to snake_case like tracker_code)

X-Paging-SortOrder: (string) ASC / DESC

X-Paging-Search: (string) Free-text search keyword

- Advance Pagination Headers:

X-Paging-Offset: (long) Start index (e.g., 0)

X-Paging-Limit: (long) Max number of records (e.g., 25)

X-Paging-SortBy: (string) Property name to sort by (e.g., trackerCode or converted to snake_case like tracker_code)

X-Paging-SortOrder: (string) ASC / DESC

X-Paging-Operator: (string) AND / OR

X-Paging-IgnoreCase: (bool) true / false

X-Paging-Include: Base64-encoded JSON filters using property names ({ "tracker_code": "REQ-ACC|REQ-CREATE"}), use "||" as the separator to search multiple values in the same field

X-Paging-Exclude: Base64-encoded JSON filters to exclude using property names ({ "tracker_code": "REQ-CANCEL|REQ-COMPLETE"}), use "||" as the separator to search multiple values in the same field

API Docs – Paginations - Example

- Simple Pagination Headers:

X-Paging-Offset= 0

X-Paging-Limit: 25

X-Paging-SortBy: id

X-Paging-SortOrder: ASC

X-Paging-Search: REQ-CREATE

- Advance Pagination Headers:

X-Paging-Offset: 0

X-Paging-Limit: 10

X-Paging-SortBy: tracker_code

X-Paging-SortOrder: DESC

X-Paging-Operator: OR

X-Paging-IgnoreCase: true

X-Paging-Include: eyDigJx0cmFja2VyX2NvZGXiJ06lOKAnFJFUS1BQON8fJFUS1DUkVBVEXigJ19

X-Paging-Exclude: bmFtZT1SZWphbA==

Thank You

